

M3U & M3U_Plus Playlist Protocol

Comprehensive Technical Specification & Reference Manual

1. Overview & Evolution

The **M3U** (Moving Picture Experts Group Audio Layer 3 Uniform Audio Locator) format originated as a basic format for listing local mp3 files. With the growth of modern IPTV and over-the-top streaming frameworks, the architecture transformed into **M3U8** (using UTF-8 string rendering) and ultimately **M3U_Plus**.

Unlike database query structures (such as the Xstream API protocol) that query specific data fragments on demand, an M3U file is a single linear plain text file. The media engine downloads the complete file locally, caching and evaluating it line-by-line to construct a directory structure inside the player application's interface.

2. Anatomy of a Basic M3U File

At its minimum configuration layer, an M3U document stack is composed of a document header row, paired with alternating media item metadata descriptors and active asset streaming targets.

Core Directives

- **#EXTM3U**: The mandatory start banner. It must be positioned on line index 1 of any valid playlist file to trigger media parsing modules.
- **#EXTINF**: The basic information row defining video duration indicators, metadata properties, and user-facing layout text descriptors.
- **The Target URI Path**: The concrete file link location or network protocol layout resource address (e.g., MPEG-TS transport streams, HLS manifests, or raw file packages).

Basic Implementation Example

```
#EXTM3U
#EXTINF:-1,Example Live Channel
http://streaming.server.com:8080/live/user/pass/101.ts
#EXTINF:120,Example VOD Trailer Movie
http://streaming.server.com:8080/movie/user/pass/502.mp4
```

Note on Playback Durations: For live broadcast streams, the duration integer parameter directly attached following the **#EXTINF**: tag is written as **-1** or **0**. This identifies the line stream as an infinite live broadcast

block, suppressing tracking calculations. Positive values like 120 describe absolute pre-recorded runtimes calculated strictly in seconds.

3. Extended M3U (M3U_Plus) Specification

To enable media application players to parse complex categorical channels, brand logos, and television electronic guide mappings, the IPTV development landscape introduced attribute elements directly inside the tracking #EXTINF string. This variant layer is defined as the **M3U_Plus** standard format.

Structural Syntax Blueprint

```
#EXTINF:<duration> <attribute1>=<"<value>" <attribute2>=<"<value>" ,<channel_name>
```

Core IPTV Attribute Matrix

Attribute Key	Data Matrix Profile	Functional Pipeline Objective
tvgl-id	String (e.g., BBCOne . uk)	The unique key code assigned to synchronize this media target string with a matching database node inside an XMLTV electronic guide file.
tvgl-name	String (e.g., HBO HD East)	The formal tracking name designator evaluated by database management operations.
tvgl-logo	URL String Address	Directs the client runtime layout engine to retrieve visual images (PNG/JPG) to display as an icon inside the player.
group-title	String Text Category	The category assignment value. Applications use this text pattern to group a list containing thousands of elements into navigation folders.
tvgl-chno	Integer Value (e.g., 101)	Forces a logical sequence position inside electronic guides, overriding alphabetical sorting logic.
radio	Boolean Code (1 or 0)	If set to 1, instructs the media layer to bypass video rendering processes and execute audio-only processing.

Production Form Syntax Example

```
#EXTINF:-1 tvg-id="CNN.us" tvg-name="CNN News HD" tvg-logo="https://server.com/img/cnn.png" group-title="US News" tvg-chno="204",CNN International HD  
http://cdn.streamprovider.net/live/user/pwd/4412.m3u8
```

4. Structural Global Headers & Advanced Controls

Global attributes can expand the #EXTM3U initialization tag sequence on Line 1. This permits the playlist provider to define system-wide behaviors affecting how clients sync with the playlist.

```
#EXTM3U url-tvg="http://epgserver.com/guide.xml.gz" refresh="3600"
```

- `url-tvg / x-tvg-url`: Directs the media engine to the source path hosting the XMLTV channel timeline documentation.
- `refresh`: Tells the application player the frequency (in seconds) to pull down a new copy of the manifest from the source server.

5. Content Type Structuring (Live, VOD, Series)

Because M3U formatting operates inside flat plain files, distinguishing content types relies on specific formatting patterns applied to duration variables, address terminations, and string names.

Live Television Formatting

```
#EXTINF:-1 tvg-id="HBO.us" tvg-logo="http://server.com/hbo.png" group-title="US  
Movies",HBO East HD  
http://premium.source.com:8080/live/user/pass/551.ts
```

Video On Demand (VOD / Movies) Formatting

VOD instances clear out EPG identifiers and specify an accurate runtime duration value measured in seconds. Target URIs trace directly to structural media files (e.g., .mp4 or .mkv blocks).

```
#EXTINF:7420 tvg-logo="http://server.com/covers/matrix.jpg" group-title="Sci-Fi  
Movies",The Matrix (1999)  
http://premium.source.com:8080/movie/user/pass/9012.mkv
```

TV Series & Episode Mappings

Multi-tiered show files are flattened by appending clean season and episode trackers (e.g., S01E01) to the item description string, paired with a categorical classification in the `group-title` field.

```
#EXTINF:2840 tvg-logo="http://server.com/covers/bb.jpg" group-title="Series: Breaking  
Bad",Breaking Bad S01E01 - Pilot  
http://premium.source.com:8080/series/user/pass/1041.mkv
```

6. Catchup TV & Time-Shift Archives

Advanced IPTV networks embed time-shift tracking parameters directly within standard item rows to expose recorded historic video archives via the player interface.

- `catchup`: Identifies the server-side catchup mode processing logic (e.g., `append`, `default`, `xc`).
- `catchup-days`: Integer tracking how many trailing days of content are safely stored in the server catchup cache buffer.
- `catchup-source`: A macro URL structure string used by the client media engine to format historical requests.

```
#EXTINF:-1 tvg-id="BBC1.uk" catchup="append" catchup-days="7" catchup-source="?utc={utc}  
&lutc={lutc}",BBC One HD  
http://streaming.server.com:8080/live/user/pass/402.ts
```

7. Custom Request Headers via Media Playback Routing

Certain stream platforms restrict resource connections using security checks that look for exact cryptographic validation flags inside incoming HTTP requests. The M3U format accommodates this via the `#EXTVLCOPT` configuration tag positioned below the tracking descriptor row.

```
#EXTINF:-1 tvg-id="PremiumSports.uk",Premium Sports HD  
#EXTVLCOPT:http-user-agent=CustomIPTVPlayer/2.0  
#EXTVLCOPT:http-referrer=https://secureauth.broadcastcdn.net/  
http://edge.broadcastcdn.net/live/stream55.m3u8
```

The embedded lines command the low-level player component to alter its outbound network handshakes, appending the mandatory user-agent credentials and referrer links required to bypass external firewall protections.